Overview

The PI will study the Hodge theory of Lagrangian fibrations at MIT, with sponsoring scientist Davesh Maulik. Hyper-Kähler manifolds are basic building blocks of compact Kähler manifolds with trivial canonical bundle, and classifying them is a fundamental open problem. By the recently proven hyper-Kähler SYZ conjecture, every known deformation type of hyper-Kähler manifolds admits Lagrangian fibrations, whose general fibers are abelian varieties of half the dimension. The project asks which abelian varieties occur as such fibers, i.e. where the associated period map lands.

The PI's thesis proves that for a non-isotrivial Lagrangian fibration of a $2n$-dimensional hyper-Kähler manifold, the monodromy is irreducible and the very general fiber has Mumford–Tate group $\mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^{n/h}$ up to isogeny, for some $h \mid n$. Equivalently, the fiber is isogenous to a product of h-dimensional abelian varieties, each *Hodge-generic* (with no unexpected Hodge classes). Moreover, $h = n$ for a general fibered member of each deformation type with $b_2 \geq 7$. The research objectives are:

RO 1. Determine which h occur, and whether $h < n$ forces $K3^{[n]}$ type, using the Donagi–Markman cubic condition to split the period map along algebraic foliations of the base.

RO 2. Study non-compact Lagrangian fibrations, chiefly the SL_r Hitchin fibration (whose monodromy the PI's methods show to be irreducible), and determine the Mumford–Tate group of its very general fiber via Lefschetz theory for spectral curves.

RO 3. Work toward the hyper-Kähler Bogomolov–Mumford conjecture (a uniruled divisor in every ample linear system), using typical intersections in the moduli of abelian varieties to find families of fibers with an elliptic factor, whose degenerations give rational curves.

Maulik, a leading expert on abelian fibrations, hyper-Kähler varieties, and Hitchin systems, will meet with the PI weekly; at MIT the PI will learn the sheaf-theoretic tools the project needs, such as perverse filtrations, Fourier–Mukai transforms, and moduli of sheaves on K3 surfaces.

Intellectual Merit

Since the SYZ theorem makes Lagrangian fibrations the gateway to classifying hyper-Kähler manifolds, constraints on their fibers bear directly on classification. RO 1 would show that decomposable Lagrangian fibrations are exactly those built from elliptic fibrations, as for Hilbert schemes of points on elliptic K3 surfaces, and would characterize the $K3^{[n]}$ deformation type Hodge-theoretically. RO 2 would give a new Hodge-theoretic invariant of the Hitchin system, a fundamental integrable system in non-abelian Hodge theory and geometric Langlands, complementing recent work on its topology and algebraic cycles. RO 3 would bring functional transcendence to bear on rational curves on hyper-Kähler manifolds, which are tied to their algebraic cycles.

Broader Impacts

The PI will organize a learning seminar on Lagrangian fibrations and Hitchin systems for Boston-area graduate students and postdocs, and seek support for an early-career workshop. Both will be open to all and aimed at researchers without local access to the area, with notes and recorded talks posted publicly. The PI will also volunteer with Boston-area math circles and competitions such as HMMT, favoring programs serving schools with limited access to advanced mathematics, and seek to mentor undergraduates through the MIT Summer Research Program. MIT's teacher-training program for new postdocs will further develop the PI as an educator. Finally, Lagrangian fibrations and Hitchin systems are integrable systems that appear throughout mirror symmetry and mathematical physics, so the results will be useful beyond algebraic geometry.